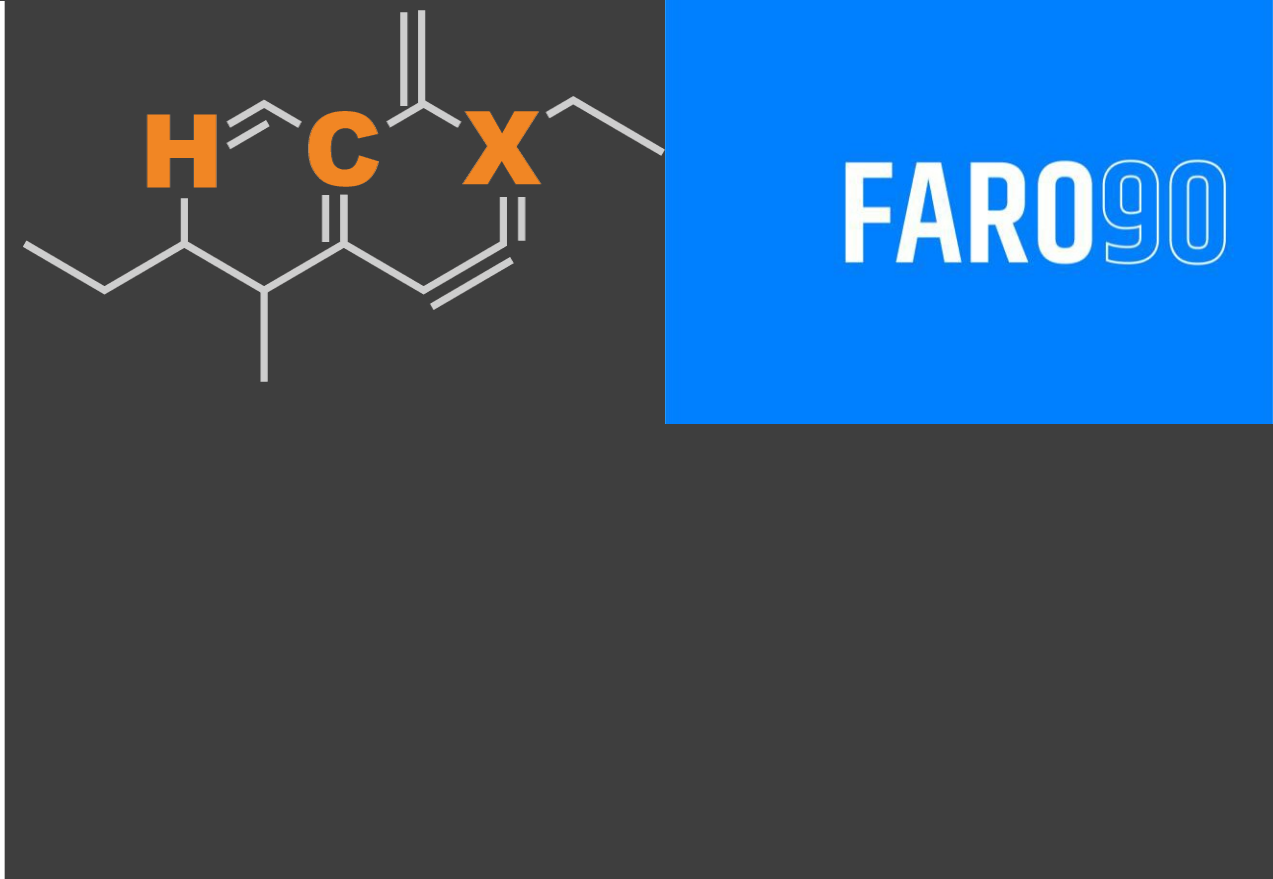




Ethanol Blending in Gasoline - Romania

September, 2025

Ethanol Blending in Europe

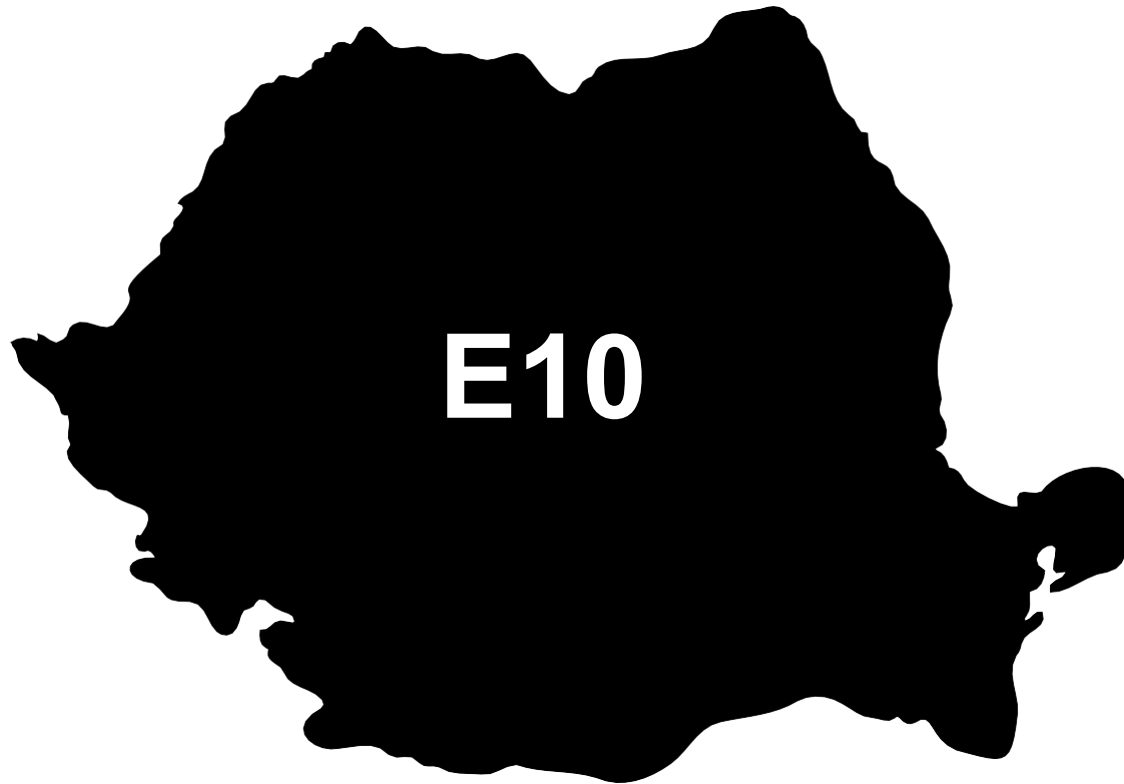
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Romania

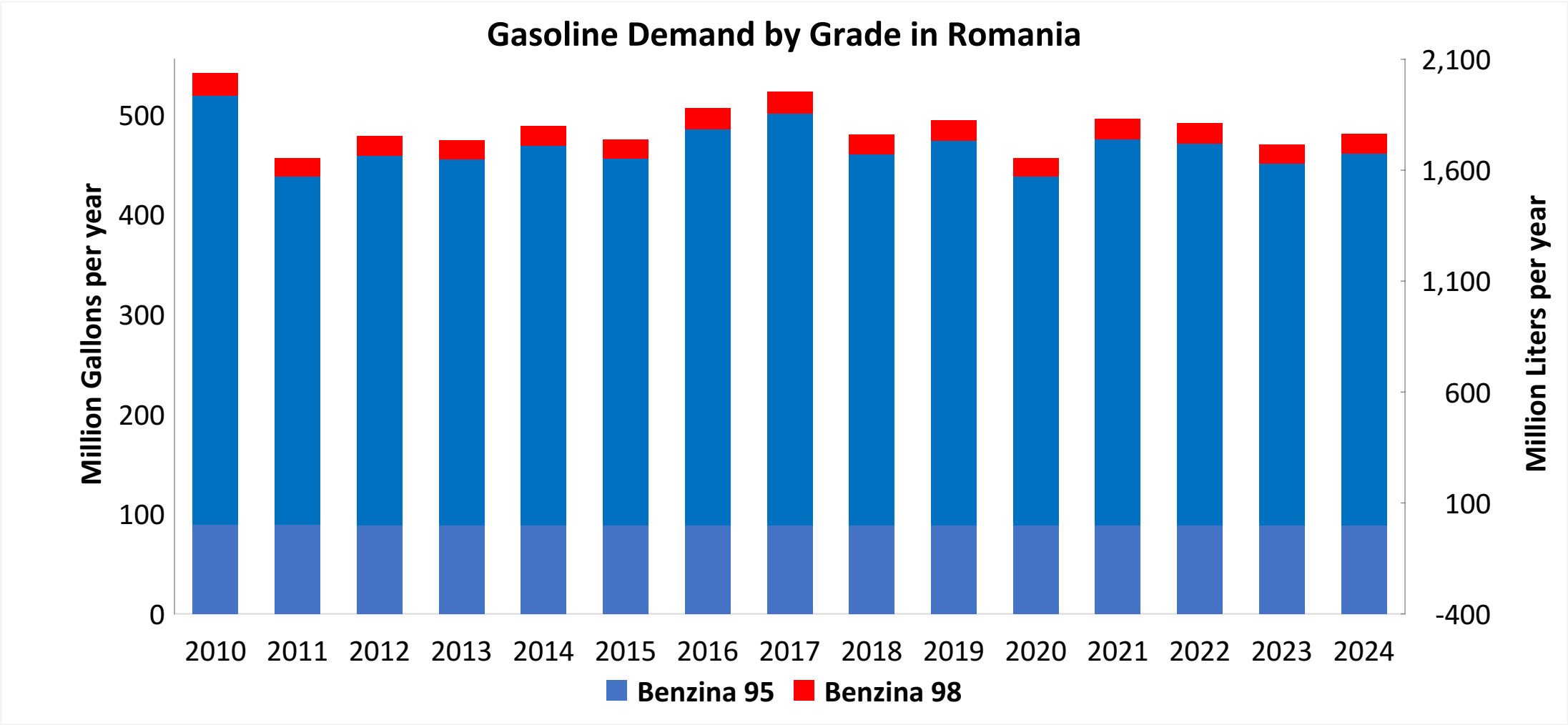


In 2024, gasoline consumption in Romania was 1,762 million liters (465 million gallons) and growth rate was -.7%. Gasoline production and net imports were 4,358 and -2,596 million liters (1,151 and -686 million gallons) correspondingly. Romania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Romania was 144 million liters (38 million gallons) representing 8.2% of gasoline demand. Ethanol production and Net Imports were 107 and 37 million liters (38 and 10 million gallons) correspondingly. Ethanol sources are Corn 28% and Cereals 72%.

Source: Eurostat, European Commission

Gasoline Demand in Romania

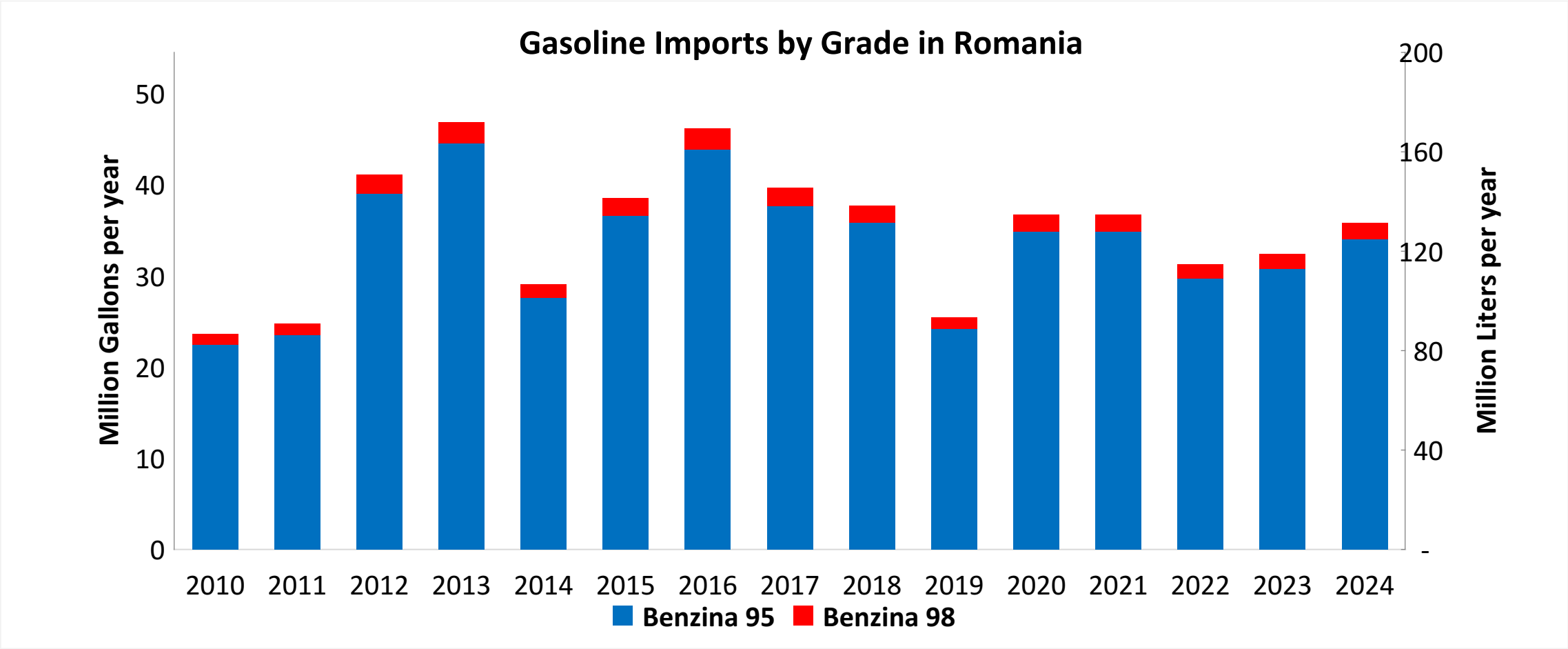


Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a cyclohexene ring with a methyl group at position 1, a vinyl group at position 2, and a substituent 'X' at position 3. The ring carbons are labeled with 'H' and 'C' in orange, and the substituent 'X' is also in orange.

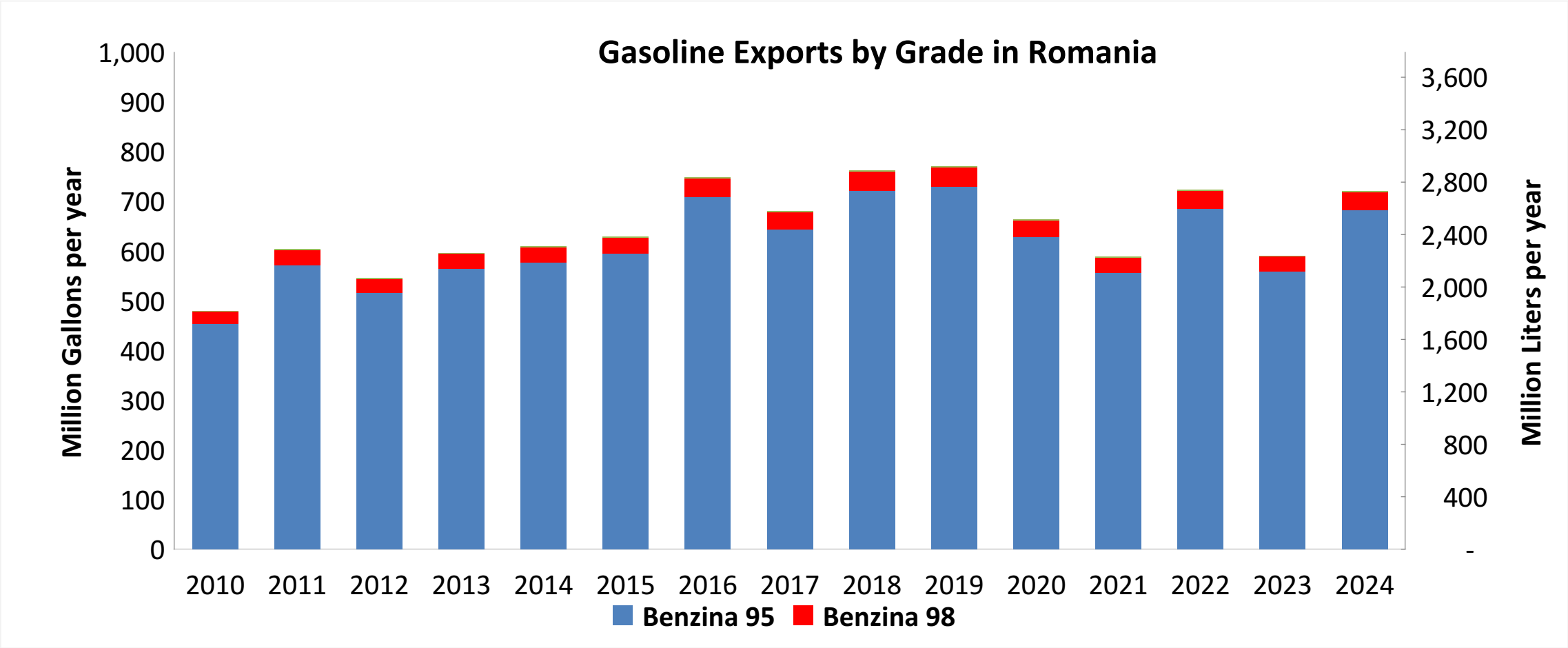


Gasoline Imports in Romania



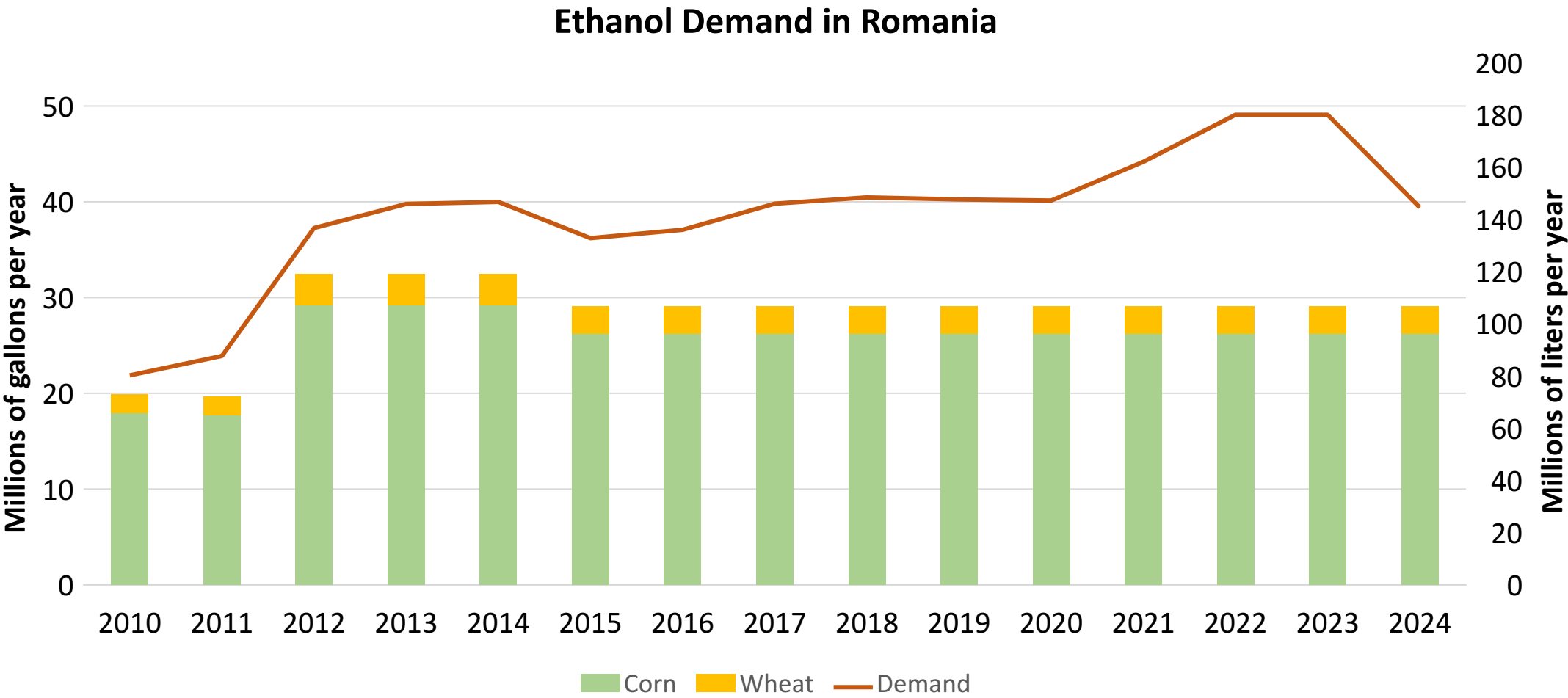
Source: Eurostat

Gasoline Exports in Romania



Source: Eurostat

Ethanol Balance in Romania



Source: Eurostat

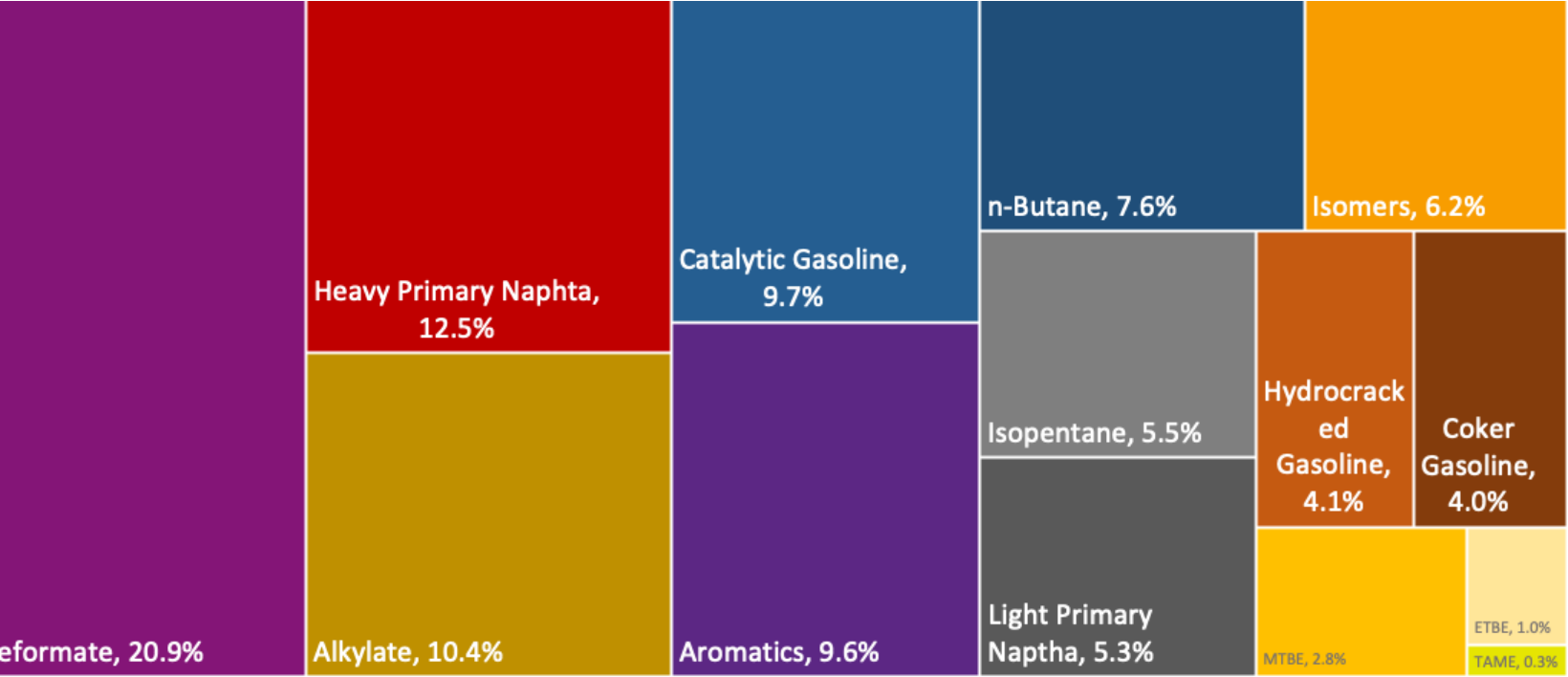
Gasoline Quality in Romania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

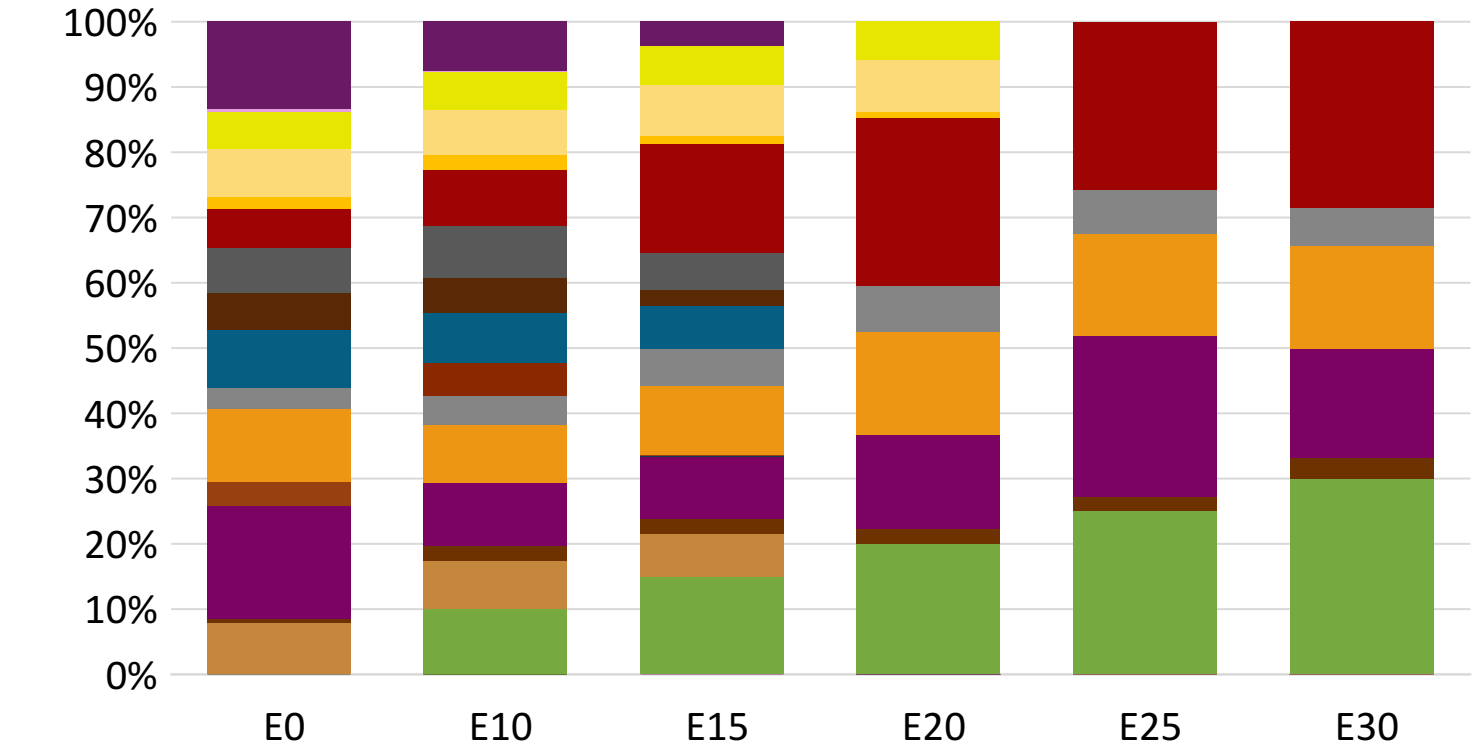
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Romania Available Blending Components



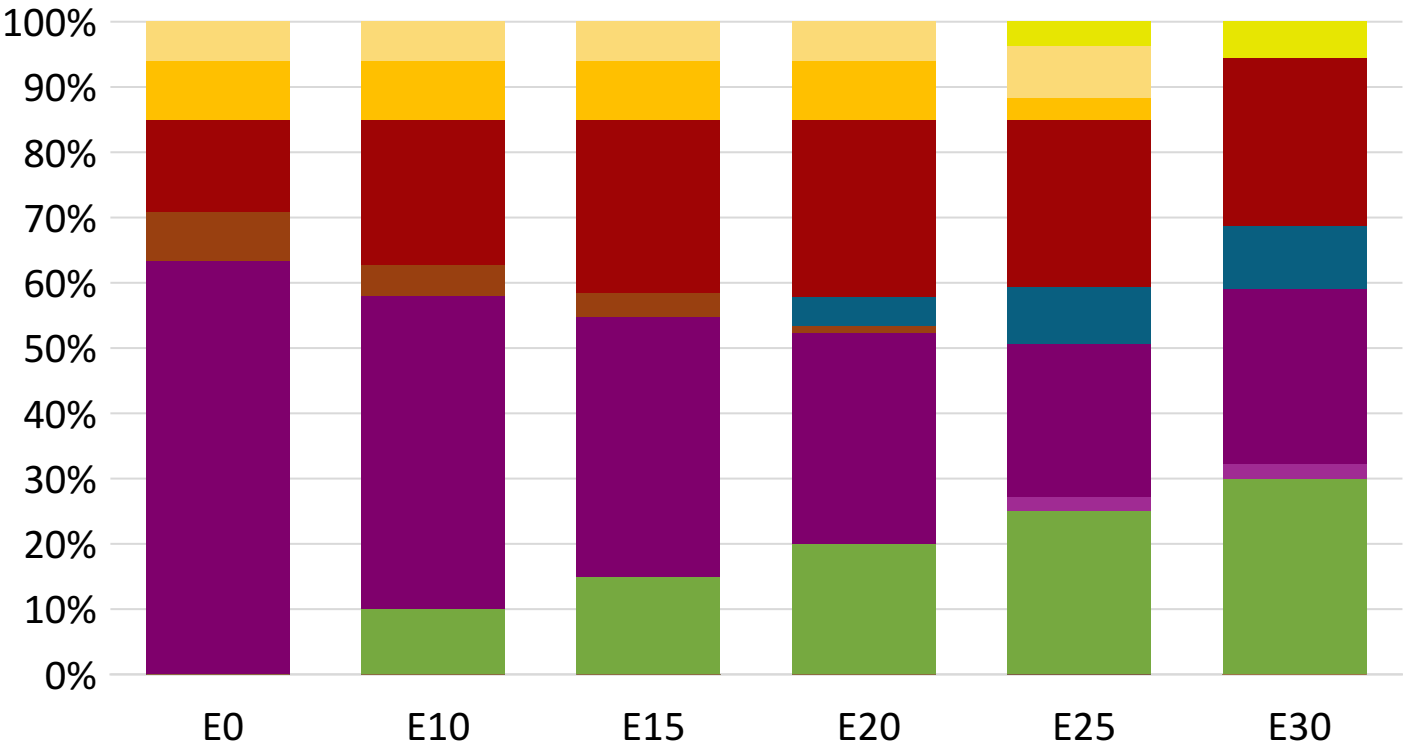
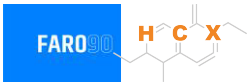
Romania – Gasoline 95 – Constant Octane



Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.284	\$ 2.146	\$ 2.046	\$ 1.931	\$ 1.885	\$ 1.832

Romania - Gasoline 98 - Constant Octane

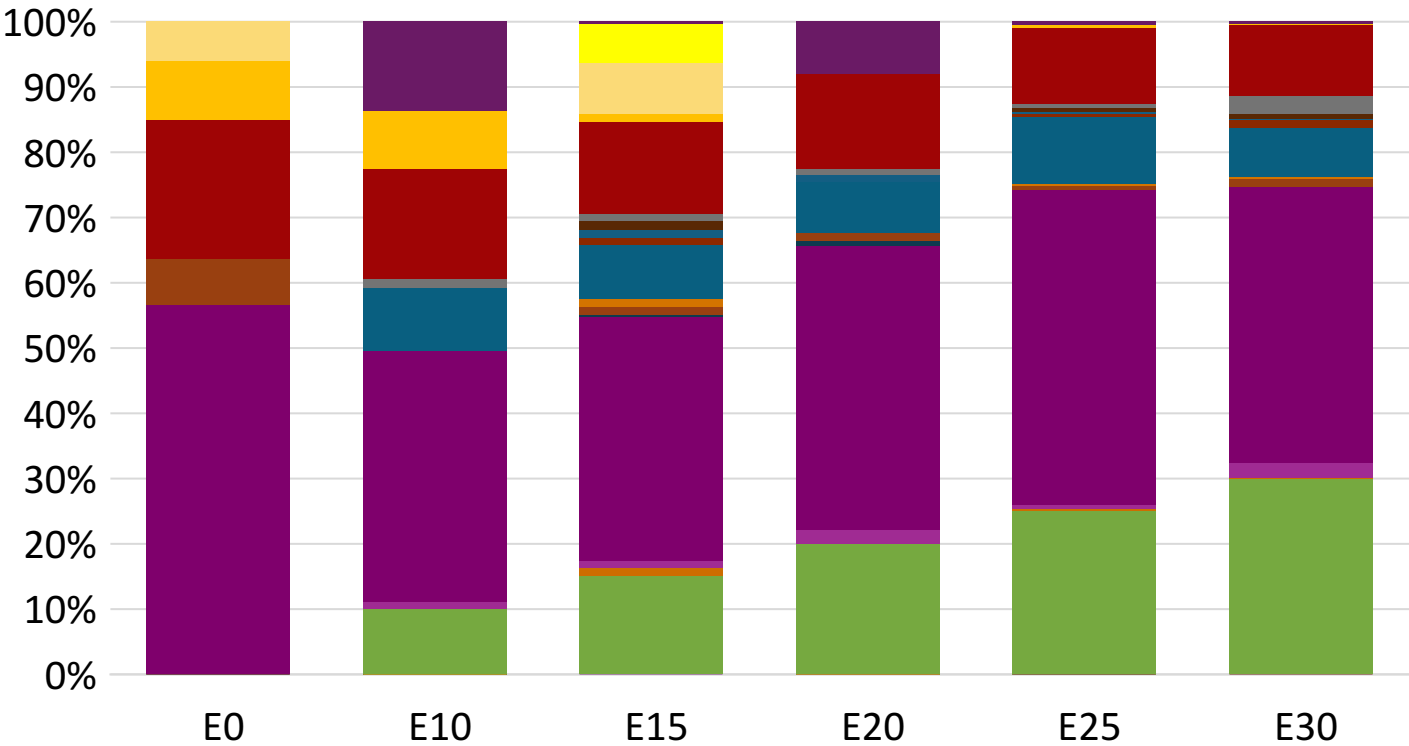


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.304	\$ 2.187	\$ 2.118	\$ 2.046	\$ 1.981	\$ 1.931

Source: Faro90

Romania – Gasoline 95 – Increased Octane

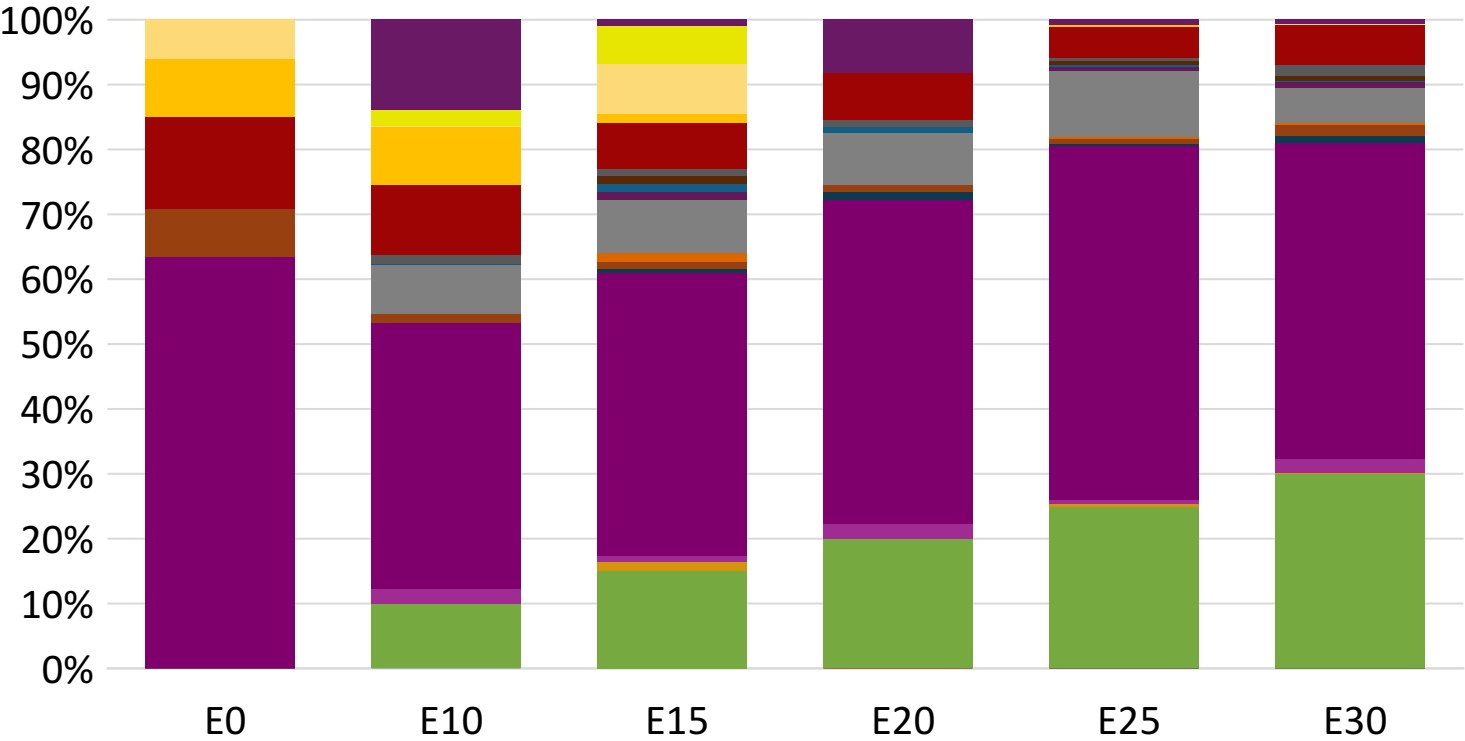


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172	\$ 2.172

Source: Faro90

Romania – Gasoline 98 – Increased Octane

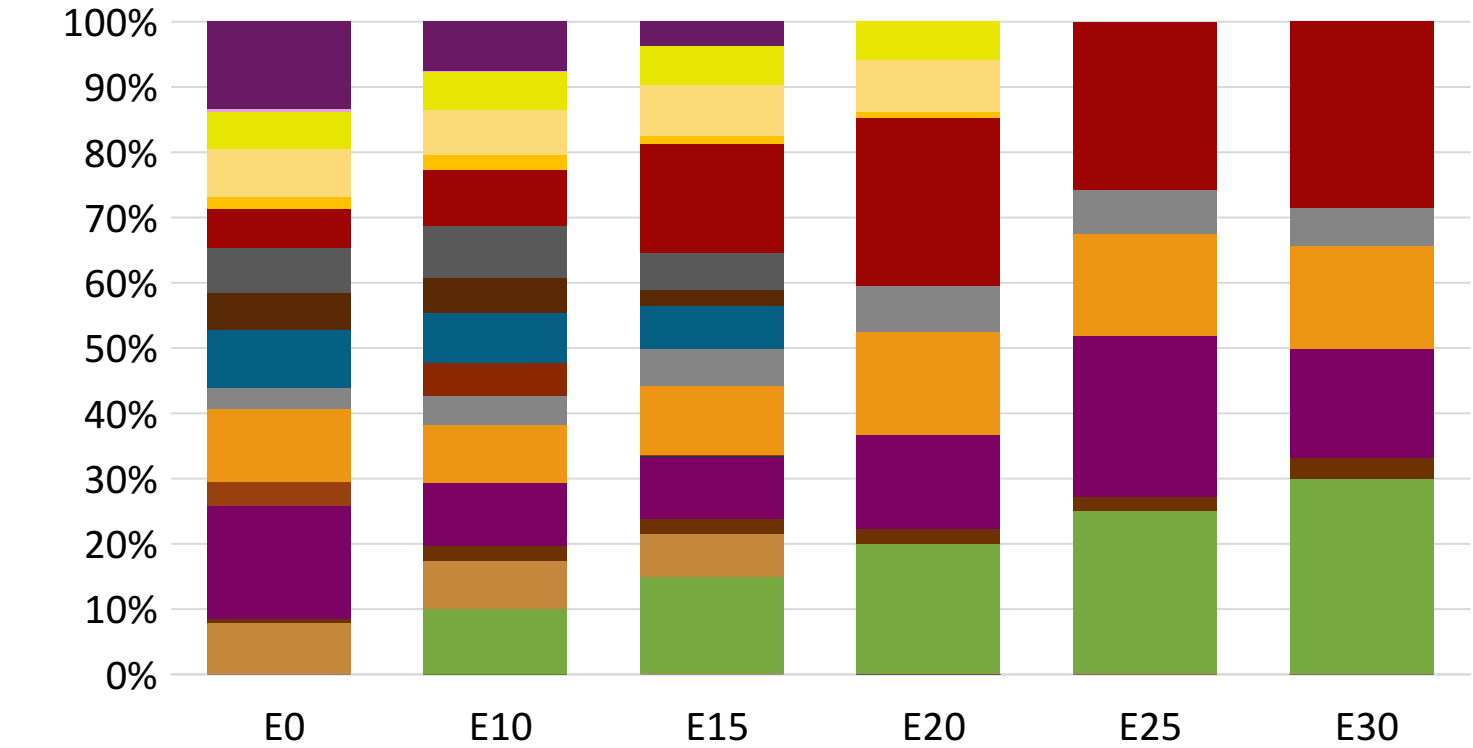


Average CIF 2024 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304	\$ 2.304

Source: Faro90

Romania – Gasoline 95 – Constant Octane



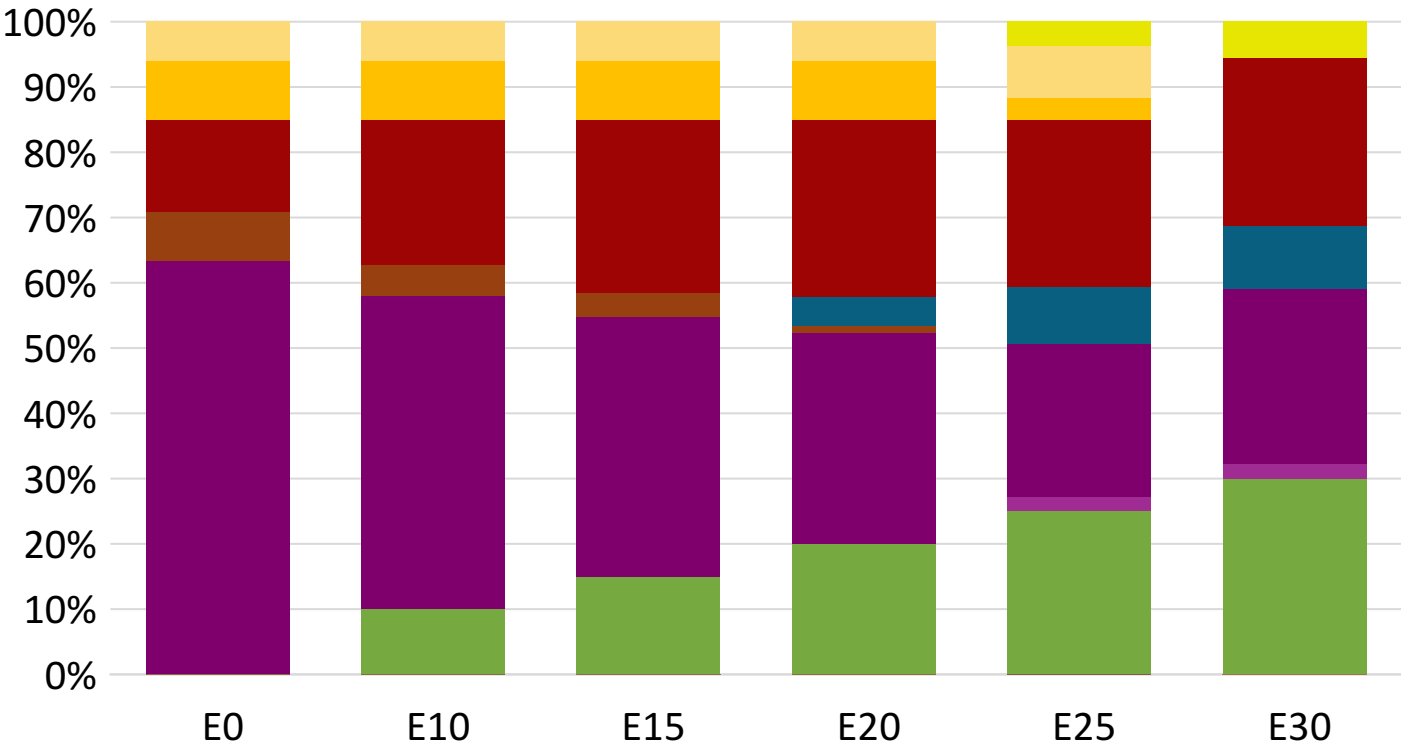
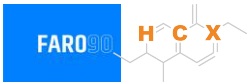
- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.274	\$ 2.136	\$ 2.036	\$ 1.921	\$ 1.875	\$ 1.822

Source: Faro90

Romania - Gasoline 98 - Constant Octane

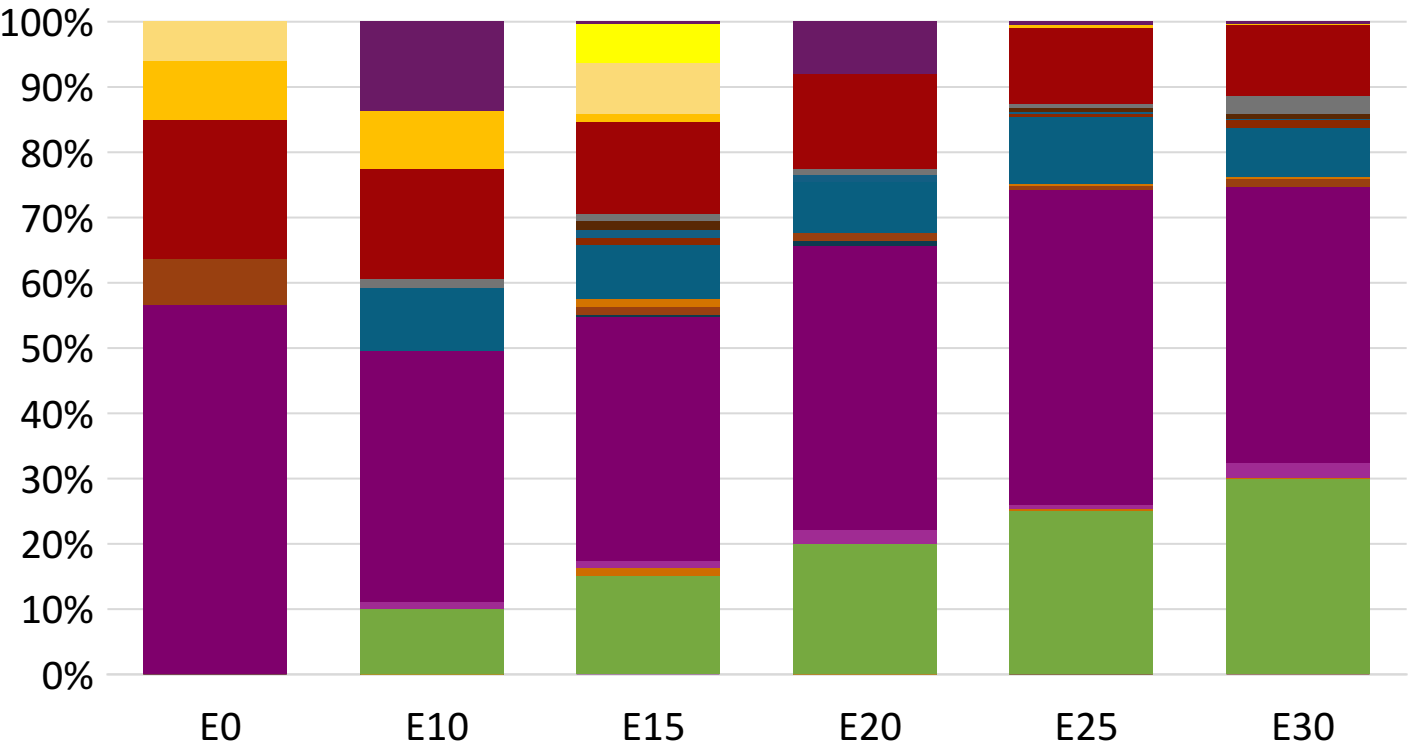


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.294	\$ 2.177	\$ 2.108	\$ 2.036	\$ 1.971	\$ 1.921

Source: Faro90

Romania – Gasoline 95 – Increased Octane

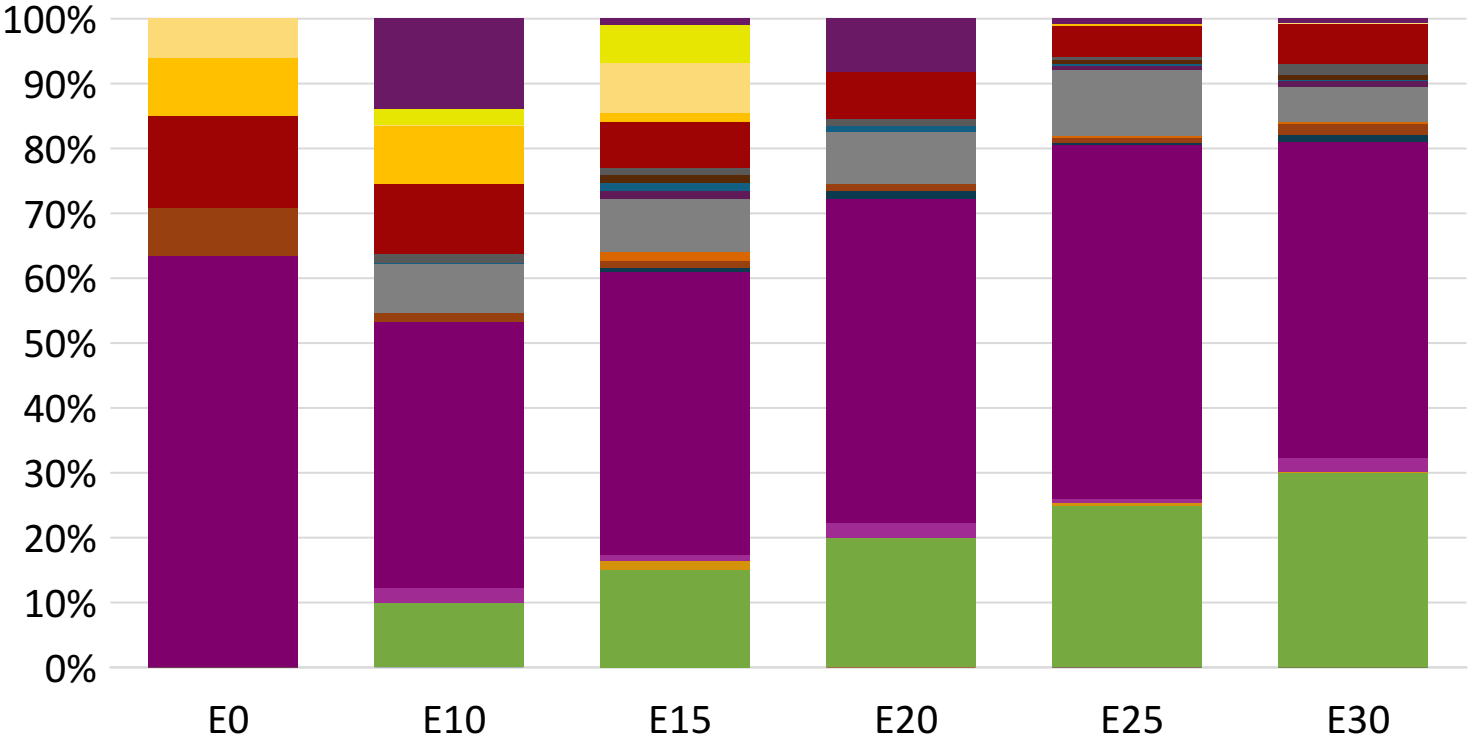


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162	\$ 2.162

Source: Faro90

Romania – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)		98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$	2.294	2.294	2.294	2.294	2.294	2.294

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

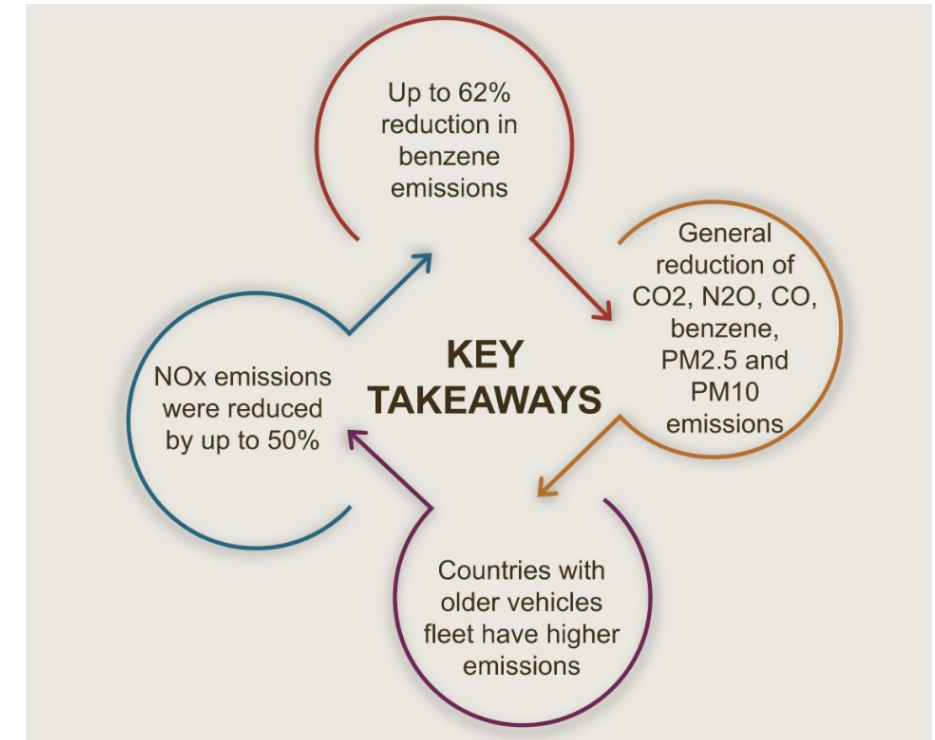
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

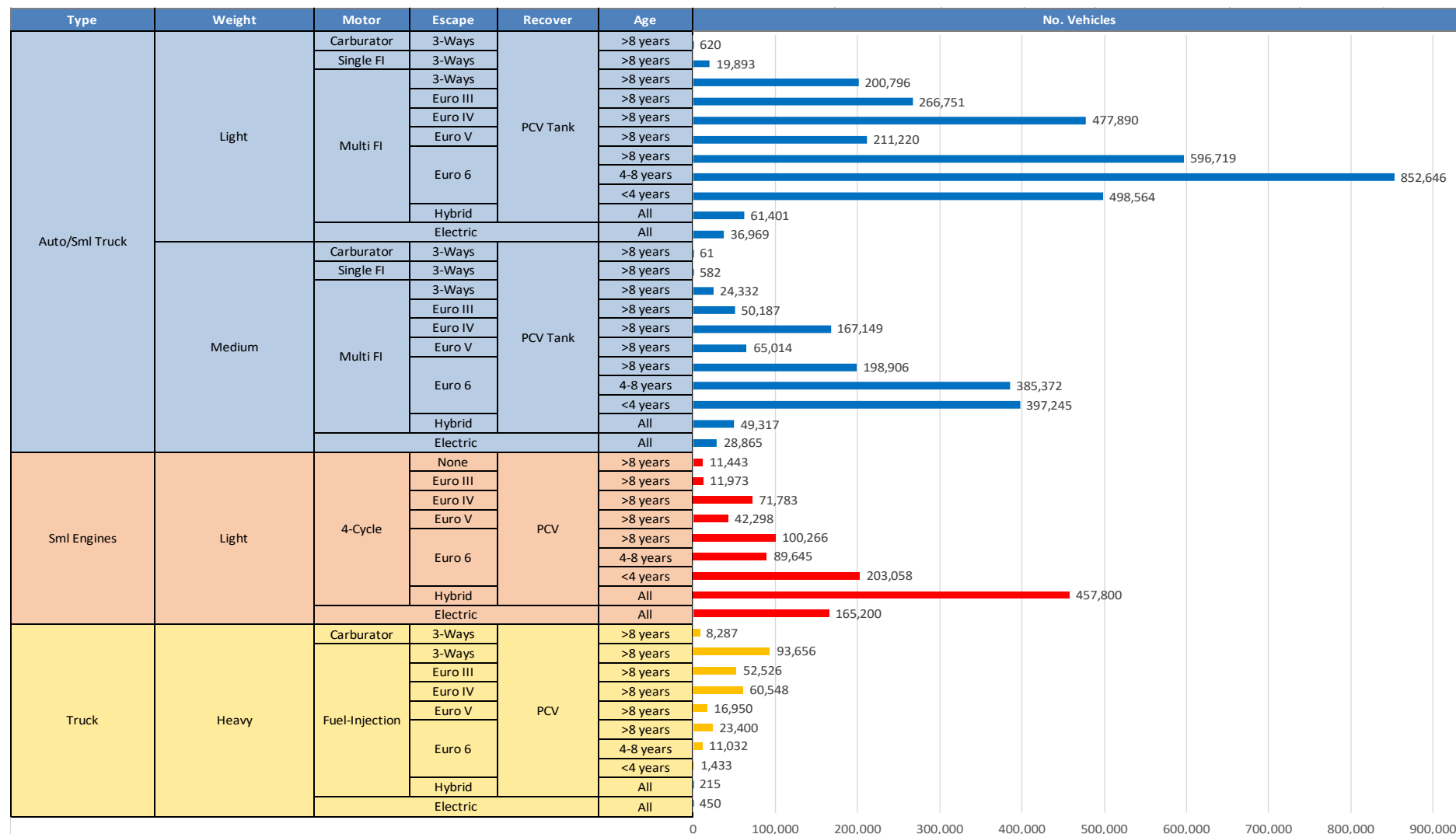
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Romania

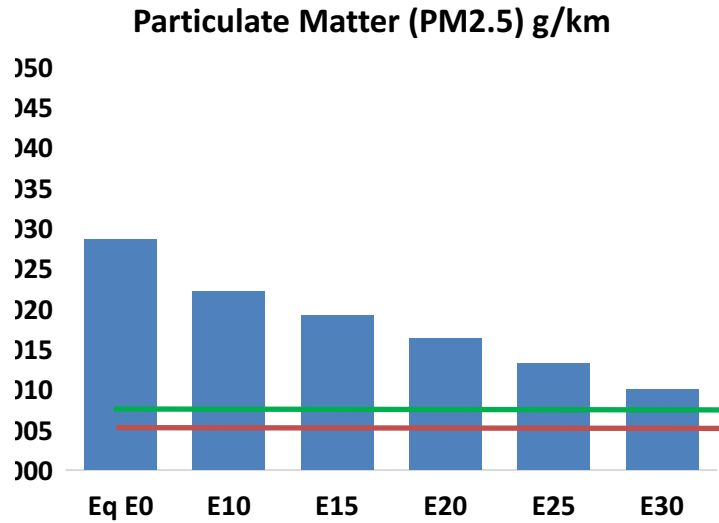
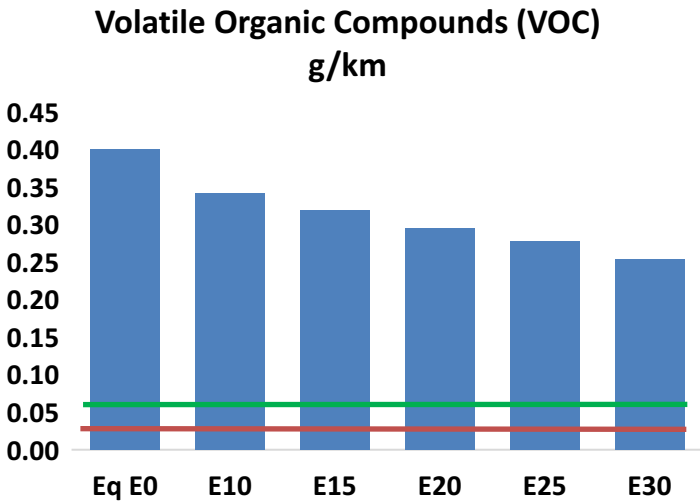
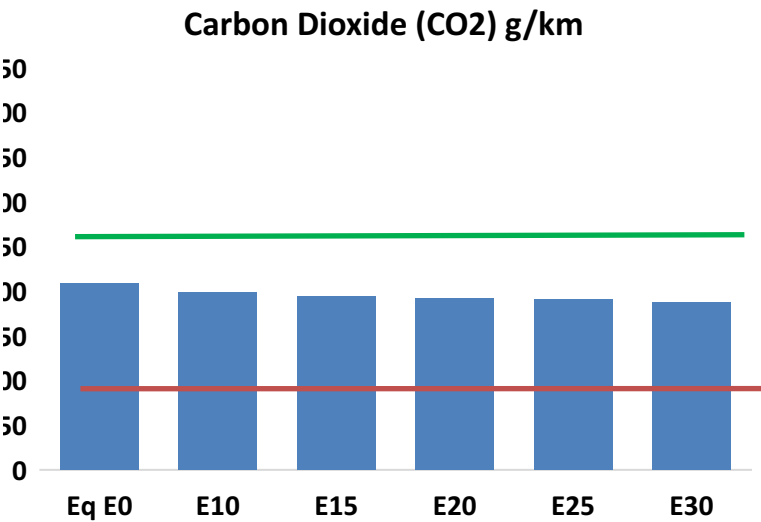
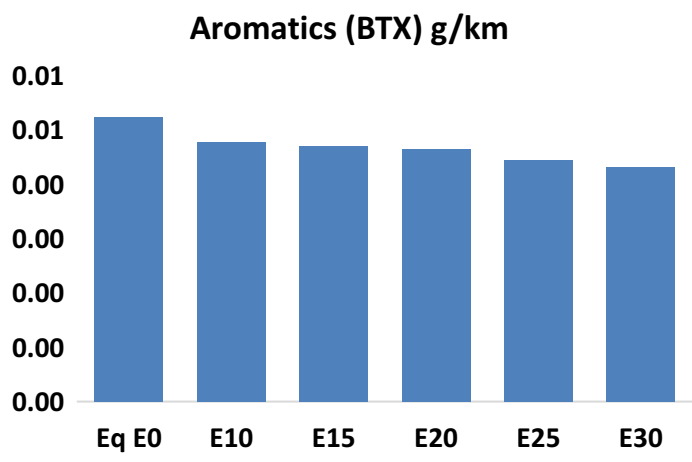
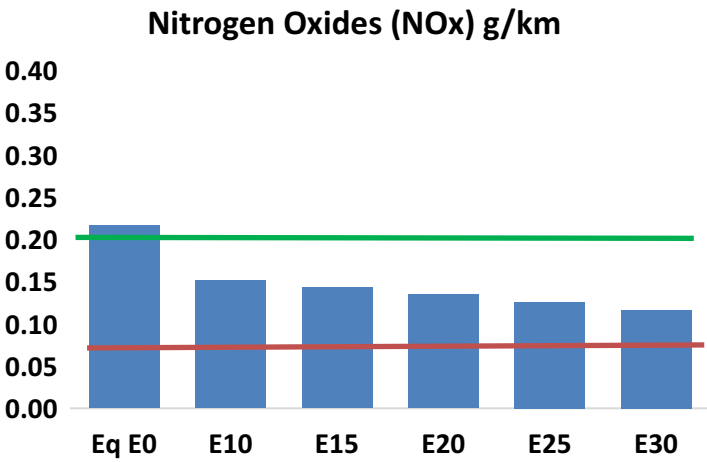
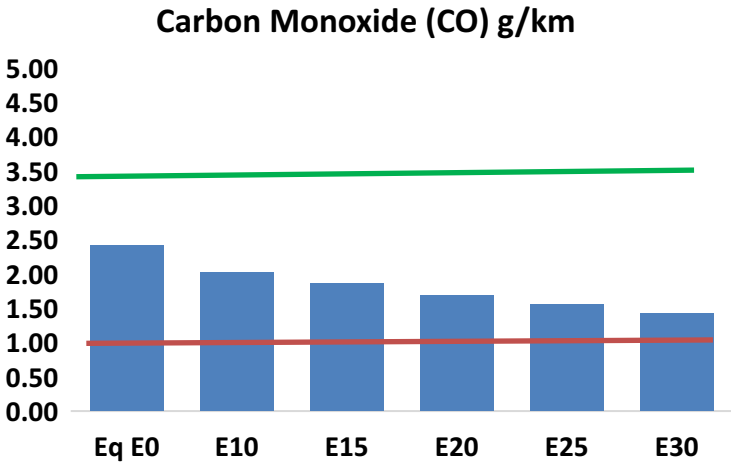
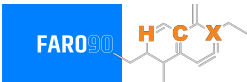


Vehicle Fleet: 6,012,464
Gasoline Vehicle Fleet: 2,699,932

Source: Eurostat, ACEA

Romania – Vehicular Emissions

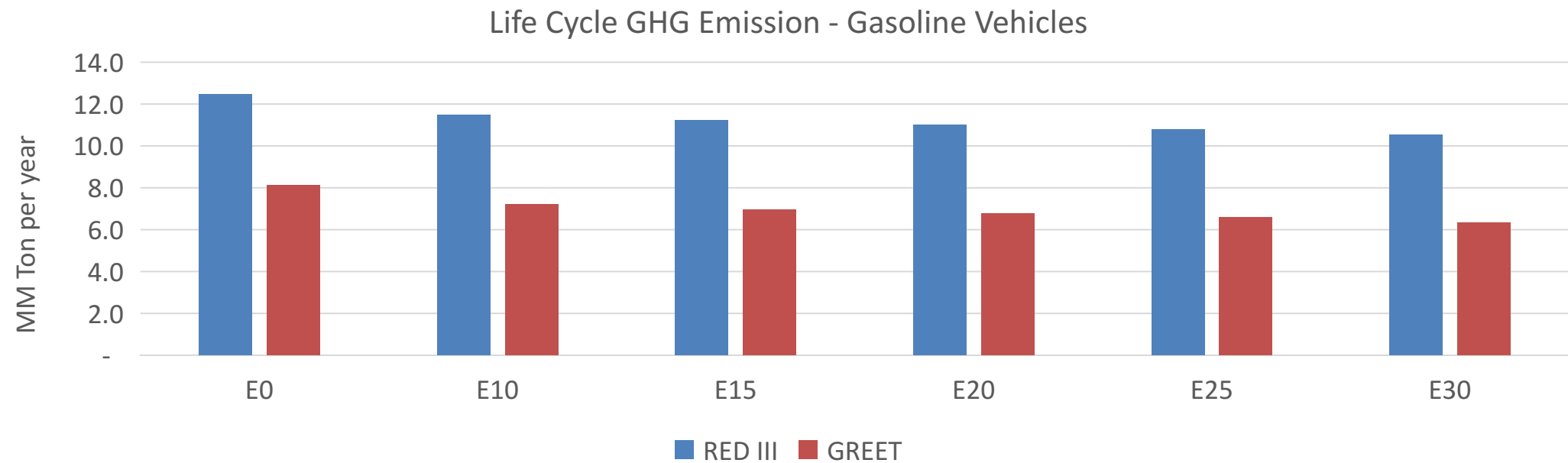
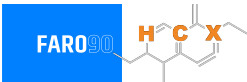
TIER III BIN 125 United States
Euro 6 Standard



Romania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	171,877	157,764	143,651	131,911	120,016	110,845	101,081	-16%	-30%
CO2	14,864,151	14,481,930	14,120,943	13,837,038	13,696,551	13,615,510	13,364,537	-5%	-8%
VOC	28,470	26,412	24,354	22,693	21,033	19,760	18,049	-14%	-26%
NOx	15,404	11,553	10,783	10,163	9,584	8,947	8,272	-30%	-38%
SOx	169	157	144	133	123	111	100	-15%	-28%
NH3	5,167	5,116	5,065	5,089	5,146	5,186	5,220	-2%	0%
N2O	247	246	245	246	251	254	257	-1%	2%
CH4	6,222	6,222	6,222	6,315	6,452	6,570	6,682	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	175	168	161	156	152	149	144	-8%	-14%
Acetaldehyde	489	561	823	1,253	1,707	1,950	2,305	68%	249%
Formaldehyde	1,766	1,717	2,002	2,350	2,462	2,724	2,965	13%	39%
Benzene	372	358	339	333	331	316	306	-9%	-11%
PM2.5	921	818	715	621	527	428	324	-22%	-43%
PM10	1,108	984	860	747	634	515	390	-22%	-43%

Romania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	76.7
Target @ 2035 (MMT/year)	43.2
Delta	33.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.3%	14.4%	16.7%	18.9%	22.0%
RED II/III	0.0%	7.8%	10.0%	11.7%	13.3%	15.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.7%	3.5%	4.1%	4.6%	5.3%
RED II/III	0.0%	2.9%	3.7%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90